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Immunostimulating activity of the polysaccharides isolated from *Cordyceps militaris*

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Abstract

In this study, the signaling mechanism of the polysaccharides isolated from fruiting body of Cordyceps militaris (CM) was investigated in macrophages to evaluate its immuno-stimulating properties. We found that CM was capable of upregulation of NO, ROS, TNF-α and phagocytic uptake in mouse peritoneal macrophages and RAW264.7 macrophages. Macrophages activation by CM seemed to occur via activation of NF-κB and all three MAPKs pathways through dectin-1 and TLR2 macrophage receptors. Additionally, we showed that CM suppressed *in vivo* growth of melanoma in an experimental mouse model. Based on these data, we suggested that CM may potentially regulate the immune response.

Research Highlights

► Signaling mechanism of polysaccharides from *C. militaris*(CM) on immunostimulation. ► Upregulation of NO, ROS, TNF-α and phagocytic uptake in macrophages. ► Macrophage activation by CM via NF-κB activation and MAPKs pathways. ► CM suppressed in vivo melanoma growth in mouse model.

Introduction

Although there are many therapeutic strategies including chemotherapy to treat cancer, high systemic toxicity to normal tissues and drug resistance has limited successful outcomes in most cases [1], [2]. Accordingly, new therapeutic strategies with improved immunity potential without harming the host are being developed to treat cancer [3], [4]. Mushrooms that have been used in traditional medicines for cancer treatment hold great promise for its use as new cancer immunotherapeutic agents. Many studies have demonstrated that polysaccharides from basidiomycete mushrooms had highly beneficial therapeutic effects including: (1) preventing oncogenesis after administration of peroral medications developed from these mushrooms or their extracts, (2) direct anti-tumor activity against various tumors, (3) synergistic anti-tumor activity in combination with chemotherapy, and (4) preventive effects on tumor metastasis [5], [6], [7], [8]. Data on polysaccharides isolated from *Cordyceps militaris* (CM) showed potential for use as cancer therapeutics. Studies have shown that CM extracts strongly suppress the growth of various tumors in animal models [9], [10].

Immunostimulation is regarded as one of the important strategies to improve the body's defense mechanism in elderly people as well as cancer patients. There is a significant amount of experimental evidence suggesting that polysaccharides from mushrooms enhance the host immune system by stimulating natural killer cells, T-cells, B-cells, and macrophage-dependent immune system responses [11], [12]. The role of activated macrophages in the defense against tumor cells has been investigated extensively. Evidence is accumulating indicating that activated macrophages are able to recognize and lyse tumor cells including those that are resistant to chemotherapeutic drugs [13]. When the body is stimulated by pathological stimuli or injury, phagocytosis is the first step in the response of macrophages to pathogens. In addition, macrophages can defend against pathogen invasion by secreting proinflammatory cytokines [*e.g.*, tumor necrosis factor (TNF)-α and interleukin (IL)-1] and releasing cytotoxic and inflammatory molecules [*e.g.*, nitric oxide (NO), reactive oxygen species (ROS)] [14], [15]. For these cellular events, various stimuli bind to pattern recognition receptors (PRRs) such as Toll-like receptors (TLRs), dectin-1 and complement receptor type 3 (CR3 or CD11b/CD18) on the surface of macrophages, and then trigger several different signaling pathways including tyrosine kinases, protein kinase C, phosphoinositide-3-kinase (PI3K)/Akt, mitogen-activated protein kinases (MAPKs), and nuclear factor κB (NF-κB) [16], [17], [18].

In this study, we evaluated that CM could enhance immunostimulating activity such as the release of toxic molecules, phagocytic uptake, and cytokine secretion in macrophages, investigated the molecular signaling pathways of macrophage activation by CM, and examined cytotoxic effects of CM on melanoma tumor growth in an experimental mouse model.

Section snippets

Materials

The crude water-soluble polysaccharides, which were obtained from the fruiting body of CM by hot water extraction and ethanol precipitation, were fractionated by DEAE-cellulose and Sepharose CL-6B column chromatography, as previously reported [19]. The purified components of CM primarily consisted of polysaccharides, which was the (1–4) or (1–2) linked glucopyranosyl or galactopyranosyl residue with a (1–2) or (1–6) linked mannopyranosyl, glucopyranosyl, or galactopyranosyl residue as a side …

Macrophage activation of CM *in vitro and in vivo*

Macrophage activation can play a role in novel immunotherapeutic approaches for the treatment of cancer [13]. Therefore, we examined whether CM was able to stimulate the functional activation of macrophages. CM effectively upregulated the macrophage functions, such as NO release, ROS generation, phagocytic uptake, TNF-α production, and morphological changes in a dose-dependent manner (Fig. 1, Fig. 2). To ensure that the effects of CM were not due to endotoxin contamination, CM was treated with …

Discussion

Cytotoxic chemotherapeutic drugs cause collateral damage to cells of the immune system and as a consequence marked systemic immune suppression [29]. Mushroom-derived polysaccharides are the most potent mushroom-derived substances in terms of their wide variety of biological activities. Therefore, mushrooms hold great promise as a potential source of new therapeutic agents [3], [4]. In the present study, we showed that intraperitoneal or oral administration of CM suppressed the growth of a …

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